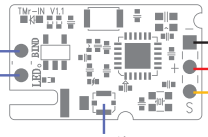
产品介绍 Introduction

FS-UBT-TMr-IN 是一款基于 AFHDS 3 协议的微型接收机。它配备单天线，支持单双向传输和回传功能。它支持 1 个 Newport 接口（Newport 接口可自定义输入或输出的信号类型：i-BUS2/S.BUS/i-BUS/PPM/PWM 等）。三防漆防护等级，用于适配机器人模型。

The FS-UBT-TMr-IN is a micro receiver based on the AFHDS 3 protocol. It is equipped with one antenna and supports both one-way and two-way transmission as well as telemetry. The receiver features one Newport interface that can be configured for various input or output signal types, including i-BUS2, S.BUS, i-BUS, PPM and PWM. With an IPC protection rating, it is compatible with RC robot models.

接收机概览 Receiver Overview



BIND 焊盘 /BIND Pad
LED 焊盘 /LED Pad
- 焊盘（电源负极）/Power Cathode Pad
+ 焊盘（电源正极）/Power Anode Pad
S 焊盘（信号端）/Signal Pin Pad
天线 /Antenna

注意：焊接时请规范操作，确保安全。焊后务必进行焊点质量检查。

Follow safe soldering practices. All solder joints must be inspected post-welding.

产品规格 Product Specifications

- 产品型号：FS-UBT-TMr-IN
- 适配模型：机器人
- 通道个数：1
- 波段个数：171
- 无线频率：2.4GHz ISM
- 无线协议：AFHDS 3
- 发射功率：<20dBm
- 天线类型：单天线
- 工作电压：3.5 ～ 9V/DC
- 通道分辨率：4096
- 数据输出：PPM/PWM/i-BUS/i-BUS2/S.BUS
- 防护等级：三防漆
- 温度范围：-10°C～ +60°C
- 湿度范围：20% ～ 95%
- 固件更新：支持
- 外形尺寸：20*12mm
- 机身重量：0.9g
- 认证 : CE, FCC ID: 2A2UNTMRIN00

- Product Model: FS-UBT-TMr-IN
- Compatible RC Models: Robots
- Number of Channels: 1
- Number of Bands: 171
- RF: 2.4GHz ISM
- RF Protocol: AFHDS 3
- Maximum Power: <20dBm (e.i.r.p.) （EU）
- Antenna: Single Antenna
- Operating Voltage: 3.5~9V/DC
- Resolution: 4096
- Data Output: PWM/PPM/i-BUS2/S.BUS/i-BUS
- Conformal Coating level: IPC protection
- Temperature Range: -10°C ~+60°C
- Humidity Range: 20%~95%
- Firmware Update: Supported
- Dimensions: 20*12mm
- Weight: 0.9g
- Certifications: CE, SRRC, FCC ID: 2A2UNTMRIN00

接口介绍 Interface Description

BIND 焊盘与 LED 焊盘： 通过分别将接收机的 BIND 焊盘和 LED 焊盘焊接引线至设备底板，即可利用设备控制接收机进入对码状态，可通过引出的 LED 状态指示灯直观了解接收机当前工作状态。

设备底板通过控制 BIND 焊盘信号来控制接收机状态：

- 设备先持续输出 1S 低电平，然后给接收机供电。供电后，立即给 BIND 焊盘输出持续高电平，此时，接收机应进入对码状态；
- 对码成功并与发射机建立连接后，设备须确保持续给 BIND 焊盘输出高电平。

设备底板则通过监测 LED 焊盘的高低电平信号，判断接收机的通信状态：

- 正常通信：LED 焊盘输出持续低电平。
- 对码中：LED 焊盘输出频率为 5Hz（周期 0.2 秒）的方波，即高低电平每隔 0.1 秒交替一次。
- 掉码 / 未连接：LED 焊盘输出频率为 0.5Hz（周期 2 秒）的方波，即高低电平每隔 1 秒交替一次。

BIND Pad and LED Pad: Solder cables from the receiver's BIND and LED pads to the mainboard. The mainboard can then force the receiver into binding mode and monitor its state via the extended LED.

The mainboard controls the receiver's state by manipulating the signal on the BIND pad:

- The mainboard first outputs a continuous low-level signal for 1 second, then powers on the receiver. Immediately after power-up, the mainboard outputs a continuous high-level signal to the BIND pad. At this point, the receiver should enter the binding state.
- After successful binding and connection to the transmitter, the mainboard must keep the BIND pad continuously high.

The mainboard determines the receiver's communication status by monitoring the high/low-level signal on the LED pad:

- Normal Communication: The LED pad outputs a continuous low-level signal.
- Binding: The LED pad outputs a 5Hz (period 0.2s) square wave, meaning the level alternates between high and low every 0.1 second.
- Signal Loss / No Connection: The LED pad outputs a 0.5Hz (period 2s) square wave, meaning the level alternates between high and low every 1 second.

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对码 Binding

FS-UBT-TMr-IN 接收机支持双向对码和单向对码。在双向对码完成后，发射机会显示接收机回传的信息。因此，对码之前，需要在发射机端先设置好是单向对码还是双向对码。如需对码接收机与发射机，请按照以下步骤操作：

双向对码步骤：

1. 发射机选择双向通信，然后进入对码状态；
2. 使接收机进入对码状态：
   - 设备先持续输出 1S 低电平，然后给接收机供电。供电后，立即给 BIND 焊盘输出持续高电平，此时，接收机应进入对码状态。
3. 当接收机的 LED 灯常亮时，表示对码成功（发射机在对码成功后会自动退出对码状态。）；
4. 检查发射机、接收机是否正常工作。如需重新对码，请重复以上步骤。

单向对码步骤：

1. 发射机选择单向通信，然后进入对码状态；
2. 本接收机进入对码状态（具体进入对码状态的方法请参考双向对码时的描述）；
3. 接收机 LED 灯变为慢闪后将发射机退出对码状态。此时，如果接收机的 LED 灯常亮，表示对码成功；
4. 检查发射机和接收机是否正常工作。如果需要重新对码，请重复上述步骤。

The FS-UBT-TMr-IN receiver supports both two-way and one-way binding. After completing two-way binding, the transmitter will display the information returned by the receiver. Therefore, before binding, it is necessary to set whether it is one-way binding or two-way binding on the transmitter side. If you need to bind the receiver with the transmitter, the steps are as follows.

Follow the steps below to bind in two-way binding:

1. Select [2 WAY] for RF standard of the transmitter, then put the transmitter into the binding mode.
2. Put the receiver into the binding mode:
   - The mainboard shall initially drive its output low-level signals for 1 second. After this interval, power shall be applied to the receiver; simultaneously, the mainboard shall drive the BIND pad high-level signals and hold it high. This sustained high-level triggers the receiver to enter binding mode.
3. When the receiver's LED is solid on, it indicates that the binding is successful (The transmitter will automatically exit the binding state after a successful binding).
4. Check if the transmitter and receiver are working properly. If you need to re-bind, please repeat the above steps.

Follow the steps below to bind in one-way binding:

1. Select [1 WAY] for RF standard of the transmitter, then put the transmitter into the binding mode.
2. This receiver enters the binding state (for details on how to enter the binding state, refer to the description provided for two-way binding).
3. After the receiver LED becomes slow flashing, then put the transmitter to exit the binding state. If the receiver's LED is solid on, it indicates that the binding is successful.
4. Check to make sure the transmitter and receiver functions are working correctly, repeat steps above(binding process) if any problems arise.

固件更新 Firmware Update

本接收机的固件更新需要通过富斯遥控管家（FlySkyAssistant）来完成。请注意，只有 3.0 及以上版本的富斯遥控管家支持此操作，相关固件可以从官网 www.flyskytech.com 下载。

步骤如下：

1. 完成发射机与接收机的对码（接收机 LED 灯常亮）；
2. 将发射机连接到电脑。在电脑端打开富斯遥控管家软件，通过该软件完成固件更新。

The firmware update for this receiver must be completed through FlySkyAssistant. Please note that only versions 3.0 and later of FlySkyAssistant support this operation, and the relevant firmware can be downloaded from the official website www.flysky-cn.com.
The update process is as followed:

1. Complete the binding between the transmitter and the receiver (the LED of the receiver is solid on).
2. Cconnect the transmitter to the computer. Open FlySkyAssistant on the computer and perform the firmware update.

失控保护 Failsafe

失控保护功能用于当接收机无法正常收到发射机的信号时，接收机按设置好的失控保护值进行通道输出以保护模型及人员安全。

本款接收机共支持三种失控保护模式：[无输出]、[保持] 和 [固定值]

[无输出] PWM 通道接口为无输出状态；

[保持] 输出失控前最后的通道值；

[固定值] 输出设置的通道值。

注：

1. 对于 PPM/i-BUS/S.BUS/i-BUS2 等总线信号类型不允许单个或其中几个通道为 [无输出] 模式，通道设置为 [无输出] 模式时，

The failsafe function is used when the receiver loses signal and is out-of-control. The receiver performs channel output according to the set failsafe value to protect the safety of the model and personnel.

This receiver supports three failsafe modes: **No output**, **Hold**, and **Fixed value**.

No output: No output for PWM channel.

Hold: Maintain the last output value.

Fixed value : Output the failsafe values that have been set for each channel.

Notes:

1. For bus signal types such as PPM/i-BUS/S.BUS/i-BUS2, a single or several of these channels are not allowed to be in No output mode. The actual signal is held at the last output value when the channel is set to No output mode.

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失控保护 Failsafe

实际信号是保持最后输出值；

2. 因 S.BUS/i-BUS2 信号信息包含失控标志位，各通道失控保护设置被失控标志位传达给后续设备，若连接的设备支持失控标志位解析，则失控后，输出各通道设置的失控保护值；
3. 对于无失控标志位的信号 PPM/i-BUS，支持设置失控时信号 [无输出] 模式。设置为 [无输出] 模式后，不管各通道失控保护如何设置，失控后各通道均为 [无输出] 模式。

2. Because the S.BUS/i-BUS2 signal information contains failsafe flag bits, the failsafe settings of each channel are communicated to subsequent devices by the failsafe flag bits. If the connected devices support the failsafe flag bit analysis, the failsafe values set for each channel are output after out of control.
3. For the signal PPM/i-BUS without failsafe flag bits, it supports the setting of the signal to No output mode in case of out of control. After setting to No output mode, regardless of the setting of the failsafe of each channel, each channel will be in No output mode after out-of-control.

注意事项：

- 使用前必须确保本产品与模型安装正确，否则可能导致模型发生严重损坏。
- 为了一切正常，请养成先开发射机再接收机通电以及先接收机断电再关闭发射机的习惯。
- 确保接收机安装在远离电机，电子调速器或电子噪声过多的区域。
- 接收机天线需远离导电材料，例如金属棒和碳物质。为了避免影响正常工作，请确保接收机天线和导电材料之间至少有 1 厘米以上的距离。
- 准备过程中，请勿连接接收机电源，避免造成不必要的损失。

Attention:

- Make sure the product is installed and calibrated correctly, failure to do so may result in serious injury.
- Normally, you must power on the transmitter and then receiver, and power off the receiver and then the transmitter.
- Make sure the receiver is mounted away from motors, electronic speed controllers or any device that emits excessive electrical noise.
- Keep the receiver's antenna at least 1cm away from conductive materials such as carbon or metal.
- Do not power on the receiver during the setup process to prevent loss of control.

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认证相关 Certification

FCC Compliance Statement

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

Warning: changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation.
If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:
-- Reorient or relocate the receiving antenna.
-- Increase the separation between the equipment and receiver.
-- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
-- Consult the dealer or an experienced radio/TV technician for help.

EU DoC Declaration

Hereby, [ShenZhen FLYSKY Technology Co., Ltd.] declares that the Radio Equipment [FS-UBT-TMr-IN] is in compliance with RED 2014/53/EU.
The full text of the EU DoC is available at the following internet address: www.flyskytech.com/info_detail/10.html

RF Exposure Compliance

This equipment complies with FCC/ISED RF radiation exposure limits set forth for an uncontrolled environment. This equipment should be installed and operated with a minimum distance of 20 centimeters between the radiator and your body.

Environmentally friendly disposal

Old electrical appliances must not be disposed of together with the residual waste, but have to be disposed of separately. The disposal at the communal collecting point via private persons is for free. The owner of old appliances is responsible to bring the appliances to these collecting points or to similar collection points. With this little personal effort, you contribute to recycle valuable raw materials and the treatment of toxic substances.

FCC ID: 2A2UNTMRIN00

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